Taylor and Francis Book Chapter

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CHAPTER 1

Behavioral Data-Driven Prediction of Suicide Risk Using Machine Learning Approaches

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CONTENTS

Suicide is a major global health issue, and identifying those at risk early can help prevent it. Our research aims to develop a model that predicts suicidal tendencies using a dataset of 1,128 records from hospitals in Bangladesh, including Dhaka Medical College, BIRDEM, and other institutions, with additional data sourced from GitHub. The dataset includes 16 factors like age, gender, sleep disorders, past suicide attempts, mental illnesses, and mood conditions. Data cleaning, missing value processing, label encoding, and feature standardization were applied for consistency. Four machine learning models—Logistic Regression, Random Forest, Support Vector Machine (SVM), and XGBoost—were tested, with Random Forest and XGBoost achieving the highest accuracy of 98.14%. These results highlight how machine learning, particularly ensemble models, can detect subtle patterns in behavioral data, helping mental health professionals identify at-risk individuals before it's too late.

1.1 INTRODUCTION

Suicide is a worldwide critical health problem that kills almost one million people per annum. Such tragic losses can be prevented through the early identification of vulnerable people. The issue of predicting suicidal behavior is, however, complicated because the subject is sensitive and many factors are involved. The proposed study will create a risk-of-suicide model based on behavioral data that is collected in Bangladesh through famous hospitals and other platforms on the internet. This study uses 4 supervised machine learning models to predict which individuals are at risk: Logistic Regression, Random Forest, Support Vector Machine (SVM), and XGBoost. This was achieved by examining the characteristics of 1,128 records with 16 features which are demographic, psychological, and behavioral features. It aims to offer mental health professionals a powerful instrument of early intervention and prevention.

1.2 RELATED WORK

Suicide is a population issue of study that scholars. Common across governments, and in the medical community, there is a high frequency overlook. According to World Health Organization 2014 according to Global Health Estimation, in 2012 the number of people with such conditions was about 803,900 suicides which form 1.4 percent of the world mental burden or over 39 million disability-adjusted life years lost. Suicide is suicidal behavior which may also be called suicidal conduct suicidal death. Suicide was ranked 15th in 2012 greatest common cause of death in the general age groups, accounting 16 percent of injury fatality and 1.4 percent of the global demise in general. WHO predicts that more than By 2020, 1.53 million people would have lost their lives due to suicide yet 10- 20 times the number of persons attempting to commit suicide [1]. The research focused on sex-specific suicide prediction models with the Quebec Integrated Chronic Disease Surveillance System (QICDSS). Logistic Regression, Random Forest, XGBoost, and Multilayer Perceptron supervised machine learning models were utilized by the authors on individual, programmatic, systemic, and community-level characteristics. SHAP was utilized to explain the models, observing the effect of features such as psychiatric visits, social deprivation, history of trauma, and regional healthcare budgets. While model performance varied by gender, the models were both more sensitive and attained higher AUC values in females. The study demonstrated that the integration of XAI and suicide prediction is able to improve transparency and stakeholder confidence, e.g., clinicians and policymakers. The authors emphasized the importance of ethical use of data and called for increased integration of interpretable AI instruments in public health decisionmaking [2]. Various efforts have been made by the government and a number of organizations to try to prevent suicides in the country, such as opening a new office, the Parliamentary Under-Secretary of State (Mental Health, Inequalities and Suicide Prevention) [3]. Suicide is mostly one of the three most important causes of death among people 15-44 years old worldwide. Suicide ranked as the fifth most important cause of death worldwide and the most important cause of death among young people. Accordingly, WHO urged the member states to create and implementational suicide prevention programs and build new or inspect existing national suicide surveillance systems [4]. Physicians & psychologists used to use written questionnaires to identify suicide risk in their patients. The responses had been translated into scales to measure the suicide risk. ”Most of the measures and methods with low predictive value,” the big research [5]. A multi-modal feature-based system of identifying suicidal ideation using social media such as Weibo. Their strategy unites the properties of the text, image, and social aspects creating a more precise detection effect than when dealing with a single feature. Their approach using a combination of data samples, like user posts, images, emojis, and social interactions, displayed a remarkable F1-score of 79.20% to demonstrate efficacy. This paper emphasizes the need to integrate multiple modalities in the accuracy of predictions in the detection of complex behavior such as suicidal ideation [6]. A feature fusion-based ensemble approach to suicidal ideation on social media detection. Their approach would be to mix textual properties, user behavior analytics, and contextual attributes, including post rates and feedback data. The combination of diverse set of features allowed the authors to increase the prediction performance significantly. Also in their research, it can be seen that ensemble models which merge the outputs by using several classifiers have better chances of capturing the different patterns in the social media data and hence the detecting power is higher [7]. A new direction as the authors merged Transformer-based models and approaches to Explainable AI (XAI), like SHAP and LIME, to understand how people with depression and suicidal tendencies behave on social networks. Their model scored high when it came to accuracy in identifying suicidal ideation and as well became transparent in its decision-making. Such explainability facilitates the credibility of the model, particularly among the healthcare professionals when they require an explanation of how the AI has made the prediction made. The intersection of deep learning and interpretability in the context of mental health is added with this study [8]. Deep learning frameworks such as Convolutional Neural Networks (CNNs) and Recurrent

Neural Networks (RNNs) to predict the presence of suicidal ideations in social media forums through posts. Through the supervised and unsupervised learning algorithms, they showed that deep learning approaches provided an efficient way to capture the more finely balanced patterns which point to suicidal thoughts. Their work indicates the possibility of monitoring social media in real-time, which can give an opportunity to implement early interventions in order to preserve life in case of signs of distress are observed when they are not too severe [9]. A deep learning combination model composed of a grid search to optimize the hyperparameters of the model, plus cross-validation as the model validation strategy. In their model they achieved much better predictions through optimal choice of hyper parameters, and ensuring model robustness against over fitting. This combination of two approaches recommends that deep learning models should be optimized to reach optimal performances in classifications related to sensitive topics like suicide ideation by using social network data. [10] Deep learning and classical machine learning models (including Support Vector Machines (SVMs) and Random Forests) have been considered to identify suicidal ideation in social media. According to their research, hybrid models of these techniques or the combination of techniques resulted in improved recall of information and overall performance, better than the results of using a single model. Their findings demonstrate the value of learning in ensembles, as several models collaborate to leverage the strengths of any given algorithm, thereby facilitating a more effective assimilation of complex behaviors [11]. Concentrated on the ensemble deep learning methods with the joint investigation of the models such as LSTMs, GRU, and CNN to identify suicidal ideation based on posts on social media sites. Their model performed better in terms of accuracy and recall due to the added features of cross-domain, such as user behavior and post content. One key point highlighted in this research is the effectiveness of ensemble deep learning techniques in capturing various details of the data, which ultimately enhances the model's ability to recognize suicidal thoughts [12]. The first issue, about suicide gesture, is that it has been criticised because it is considered to be vague and may cause confusion when applying it in the clinical setting. They suggested different terms, including self-harm with low intent or ambivalent self-harm, to ensure that behaviors, which are frequently described as suicide gestures, can be understood clinically with more specific and accurate terms. This proposal has been practical to clarify the categories of self-harming behaviors and their part in suicide risk [13]. The Interpersonal Theory of Suicide, which suggests that the risk to commit suicide is motivated by three main factors, namely thwarted belongingness (a lack of sense of belonging to people), perceived burdensomeness (feeling that one is a burden on others), and acquired capability (the fearlessness that one has developed due to one being exposed to painful situations several times). According to the theory, these elements present together allow for an increase in suicide risk. In contrast, the factor that adds to the others needed to transform suicidal thoughts and ideas into an actual attempt is an acquired capability. Such theory has impacted the modern suicide risk assessment and disengagement techniques [14] [15].

This study seeks to determine a machine learning approach to suicide risk prediction by using a set of 1128 behavioral records associated with significant medical centers in Bangladesh. The research aims to discover the presence of key factors (age, gender, mental health conditions, past suicide attempts) and test the quality of machine learning models (Logistic Regression, Random Forest, SVM, and XGBoost) to predict suicidal behavior correctly. The final objective is to develop an effective, evidence-based instrument that would help medical workers to apply early interventions and prevent suicide.

1.3 VALUE OF DATA

A specific type of behavioral data is collected from various reliable sources, including four leading medical institutions, self-reported results in Google forms, and GitHub data curation. Thanks to such diverse data, scholars should be able to embrace varied demographic and psychological aggravations of suicidal inclinations. The data consists of such valuable variables as age, gender, civil status, occupation, education level, mental disorders, and behavior, including isolation, anger, sleep complications, and alcohol consumption. These characteristics are essential to elucidate the convoluted risk factors of suicidal ideation and behavior, particularly in developing nations, where psychopathology remains understudied. The data will enable the building and evaluation of different machine learning models to predict suicide. It can assist researchers and package developers to test new algorithms and compare the performance of various predictive models, exemplifying Logistic Regression, SVR, XGBoost, and Random Forest Regression, since it provides high-quality and real-life data. Our dataset is particularly practical for those working in the fields of public health, mental health counseling, and policymaking. The information obtained with the aid of analysis of this data may be used to form specific programs of intervention, strengthen the possibility of early identification of at-risk individuals, and, eventually, lay the foundation of high-quality suicide prevention strategies.

1.4 DATA DESCRIPTION

The current Suicide Risk Prediction Dataset contains behavioral data collected through various methods. The findings were collected from four major medical institutions in Bangladesh, which include Dhaka Medical College and Hospital, Bangladesh Medical University (BMU), Enam Medical College and Hospital, and Shaheed Tajuddin Ahmad Medical College.

Table 1.1 Source of Suicide Risk Prediction Dataset

Institute Name	**Dataset**
Dhaka Medical College & Hospital	38
Bangladesh Medical University	19
Enam Medical College & Hospital	27
Shaheed Tajuddin Ahmed Medical College	36
Google Form	358
Github	650
Total	**1128**

Table 1.2 Description of Features in the Suicide Risk Prediction Dataset

Feature	Description
Age	The age bracket that the individual belongs to (e.g., 15–24, 25–34, 35–44, etc.).
Gender	Indicates whether the individual is male or female, as mental health and behavior may differ by gender.
Religion	The religious affiliation of the person (e.g., Islam, Hinduism, Christianity, etc.).
Occupation	The main job or work type of the individual such as student, housewife, farmer, government employee, professional (doctor, engineer), unemployed, pensioner, etc.
Civil Status	Whether the individual is married, unmarried, in a relationship but not married, or a widow/widower.
Education Level	The highest level of education completed, ranging from no formal education to university degrees.
Psych Session	Indicates whether the individual has undergone any psychological or counseling sessions.
Past Attempt	Shows whether the person has a history of previous suicide attempts.
Disorder	Indicates whether the individual has been diagnosed with bipolar disorder, depression, or anxiety.
Alcohol	Whether the individual consumes alcohol regularly, rarely, or not at all.
Anger	Shows whether the individual experiences frequent or uncontrolled anger.
Sleep Problem	Indicates whether the individual suffers from sleep issues such as disturbed sleep or lack of sleep.
Socialization / Isolation	Indicates whether the person avoids social interaction, which may relate to loneliness or depression.
Humiliated	Shows whether the person feels insulted or disrespected, causing psychological harm.
Sad / Weary	Indicates feelings of sadness, depression, emotional exhaustion, or numbness.
Attempted?	Shows whether the person has attempted suicide before, identifying high-risk individuals needing urgent support.

Our dataset consisted of data from four medical institutions, collected through an online survey conducted between February 2025 and July 2025. Additionally, an opinion survey conducted via Google Forms was used to gather responses, and open-source data was analyzed on GitHub. Table 1.1 presents our primary dataset sources.Table 1.2 shows the description of the dataset. The sample of external data has been carefully selected and combined to encompass hospital records and represent individuals from diverse backgrounds. In conjunction, the primary dataset can form a strong foundation for performing risk factor analysis associated with suicidal behavior and developing predictive machine learning models. All information was gathered to correctly reflect the variation in the mental health condition of patients with different demographic and health care backgrounds. The data were collected at four prominent medical institutes in Bangladesh, namely Dhaka Medical College and Hospital, Bangladesh Medical University (BMU), Enam Medical College and Hospital, and Shaheed Tajuddin Ahmad Medical College. The record of each patient includes 16 structured features of demographic, behavioral, and psychological factors. These are: Age, Gender, Religion, Occupation, Civil Status, Level of Education, Psychological Session, Past Attempt, Disorder, Illness, Alcohol, Anger, Sleep Problem, Isolation, Humiliation, and Sad/Weary. Attempted? is the dependent variable and is a binary indicator that indicates whether the person had a history of suicide attempts. To be classified, the data have been organized into two groups of outcomes based on the level of risk. The collection of all data was conducted by all ethical considerations, and the working mental health professionals at BMU, Dhaka Medical College, and the other involved institutions provided suggestions and offered supervision.

1.5 METHODOLOGY

To determine the key indicator features for predicting the suicide rate and to identify the most suitable machine learning technique for assessing the correct suicide risk, the complete methodology is outlined in Figure 1.1.

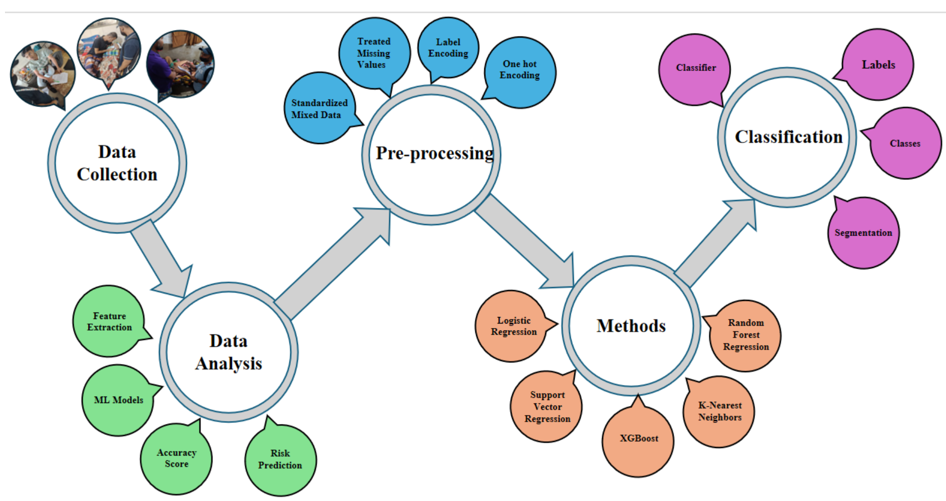

Figure 1.1 Flowchart for Suicide Risk Prediction Using ML

1.5.1 Data Collection

Our list of suicide risk factors was seen in four various medical institutions, in an online survey on Google Forms, as well as in other relevant sources on GitHub. The medical information consisted of data from hospital records and notes of counseling sessions, with sufficient anonymization to ensure privacy and confidentiality. A Google form was distributed among a diverse sample of the participants to receive self-reported data concerning mental health, suicide attempt history, psychological sessions experience, and other factors. The extracted data were supplemented with additional, filterable data on similar variables using the GitHub datasets. The geographical coordinates of the dataset collection sites are as follows in Table 1.3.

Table 1.3 Source and Location of Suicide Prediction Dataset Collection

Institution Name	**Address**	**Latitude**	**Longitude**
Dhaka Medical College & Hospital	Secretariat Road, Dhaka 1000, Bangladesh	23.727658863	90.402874016
Bangladesh Medical University (BMU)	Shahbagh, Dhaka 1000, Bangladesh	23.738745722	90.394931274
Enam Medical College & Hospital	9/3 Parbotti Nagar, Thana Road, Savar, Dhaka 1340, Bangladesh	23.838349366	90.264354229
Shaheed Tajuddin Ahmad Medical College	Bhawal Mirzapur, Gazipur 1700, Bangladesh	23.999417274	90.419622165

All data were cleaned, anonymized, and standardized after collection by excluding incomplete entries, duplicate records, and inconsistencies to provide a high-quality dataset for training and evaluating the predictive models employed in the present study.

1.5.2 Data Analysis

To derive substantial information from the collected dataset and to guarantee that the input data can be used in predictive modeling, data analysis was conducted. We explore to discover the underlying patterns, relations, and distributions of the variables that may predispose the act of suicide. The first assessment of the dataset identified the balance of data in the classes to display the rate of the population that has tried to commit suicide and those who have not. To take a closer look at the binary target variable, 'Attempted? (yes or no)', A distribution plot, shown in Figure 1.3, was produced, and it will be clear that rich models are required that can cope with possible class imbalance. To identify interdependencies across the features, a correlation heatmap was generated. Figure 1.2 shows the visualization of suicide risk. The heatmap illustrates the coefficients of pairwise correlations for all 16 structured variables. This visualization can help identify highly connected variables that contribute to the development of suicidal behavior risk, such as the connection between mental disorders and sleep difficulties, or anger difficulties and isolation. These observations helped identify pertinent characteristics and informed further preprocessing decisions to prevent multicollinearity.

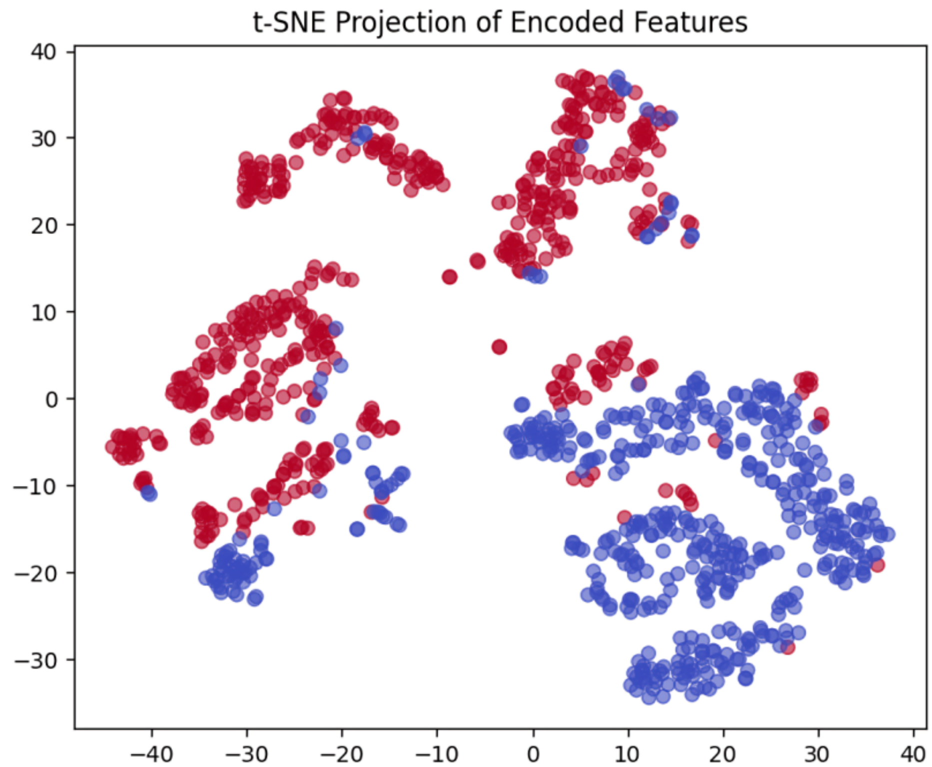

Figure 1.2 t-SNE Visualization of Suicide Risk

1.5.3 Data Pre-Processing

In the final dataset, there are 16 columns in total, where 15 columns are the indicator variables, and the Suicide risk is the target column to predict the suicide risk.

1. **Standardized Mixed Data:** Equal items that vary in spelling or format were standardized so that they are not put in the wrong group when analyzed or modeled.

2. **Treated Missing Values:** Missing (null) values were handled by replacing them with default values (e.g., mode) or removing them if irrelevant.

3. **Applied Label Encoding:** Categorical text attributes (e.g., Gender, Religion) were transformed into numerical format.

4. **One-hot Encoding:** Categorical attributes (e.g., Occupation, Civil Status) were converted to binary vectors.

5. **Removed Unnecessary Columns:** Irrelevant or duplicate columns were dropped to improve model performance.

1.5.4 Models

The study used four machine learning models to determine the possible risk of suicide using 16 chosen characteristics. Such models were selected because they are effective in the classification and regression tasks on complex behavioral data. These models include Logistic Regression, which is an easy and strong probabilistic tool, Support Vector Regression (SVR), known to deal with nonlinear relations, XGBoost, which is a robust and efficient gradient boosting algorithm that classifies based on feature similarity and Random Forest Regression, where multiple decision trees are combined to enhance prediction accuracy and eliminate overfitting. Collectively, such models would enable a comparative study to identify the most suitable model, leading to more accurate suicide risk prediction.

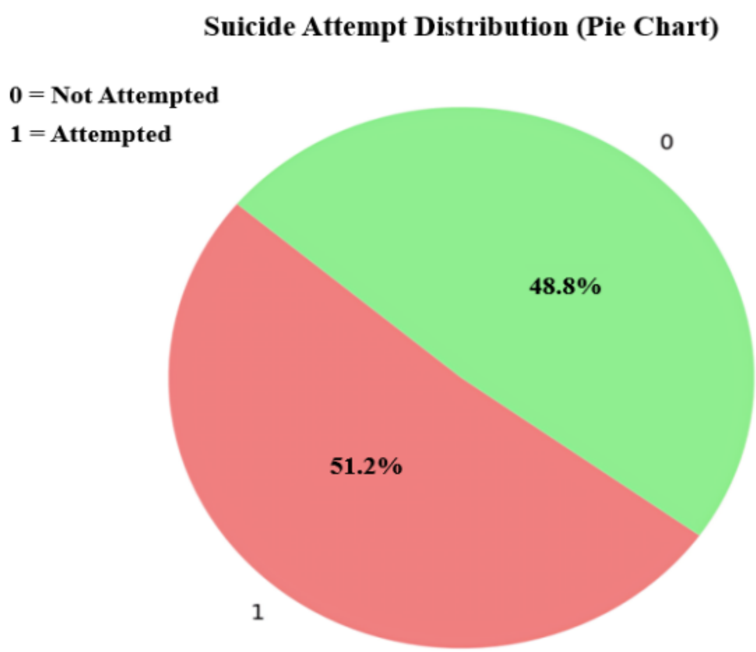

Figure 1.3 Suicide Attempt Distribution

1.6 RESULT & ANALYSIS

his paper was aimed at experimenting on four supervised machine learning models, including Logistic Regression, Random Forest, Support Vector Machine(SVM), and XGBoost, to predict potential suicidal tendencies.

We determine the four best machine learning algorithms to solve the classification problem. Four of the most well-known models were trained and tested using the dataset: Logistic Regression, Random Forest, SVM, and XGBoost. Our research explores the functioning of various classification models and was appraised. Logistic Regression is one of the most popular and easiest-to-implement binary classifier algorithms, and this algorithm estimates the probability of being in a class using the logistic function. It achieved an accuracy of 95.81%, which was not insufficient and was suitable for linearly separable data. Table 1.4 shows the different accuracy models. It is straightforward and quick, and thus ideal to use as a baseline model. On the one hand, Random Forest combines the outcomes of many decision trees to produce its final output. It provided the most significant level of precision of them all, with a peak of 98.14%. Its capabilities in dealing with overfitting and handling any complicated feature interactions make it worthwhile, especially when dealing with more complex data. Similarly, the Support Vector Machine (SVM) aims to determine the optimal separating line between classes and achieves an accuracy level of 95.81 percent, matching that of Logistic Regression. SVM particularly shines when the classes are separable, and its performance is inherently robust, which achieves stable and reliable performance. XGBoost is another advanced method that utilizes a boosting technique and achieves an outstanding accuracy of 98.14%, which is comparable to that of Random Forest. Since XGBoost can work with imbalanced data and regularization is applied to minimize the probability of overfitting, it is particularly well-suited for working with structured or tabular data sets.

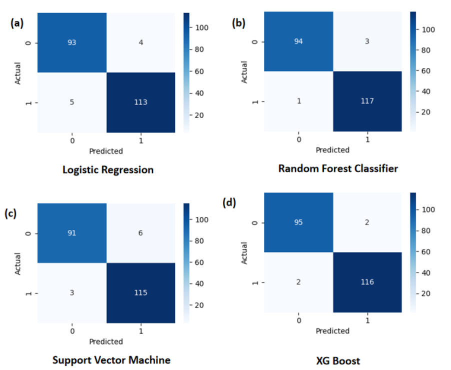

Figure 1.4 Confusion Matrix of Models

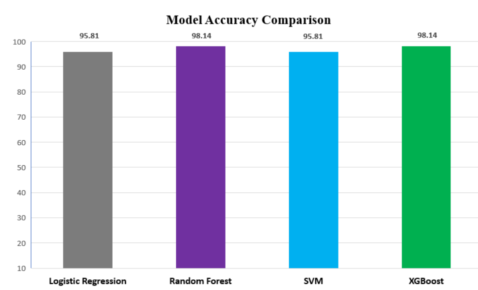

Figure 1.5 Model-wise Accuracy Evaluation

Table 1.4 Performance Metrics of Different Machine Learning Models

Model	Precision	Recall	F1-Score	Accuracy
Logistic Regression	0.96	0.96	0.96	0.958
Random Forest	0.98	0.98	0.98	0.981
SVM	0.96	0.96	0.96	0.958
XGBoost	0.98	0.98	0.98	0.981

Study results indicate that the models were successful in their execution as they scored above 93 percent in terms of accuracy. Confusion matrix of all models indicated in Figure 1.4, The fact that these models are based on a series of interrelations and interactions of the variables allowed them to work well in capturing complex relationships and interactions of variables and hence may have resulted to their high capability in predicting the rates of suicides. Figure 1.5 shows the prediction of the accuracy model.

Recall is important in suicide-risk prediction since a false high risk case omission can be disastrous. False negatives can be concealed by accuracy, and therefore a model needs to be focused on the correct identification of real at-risk individuals. Recalling stress is a way of making sure that it is a safer and more reliable method of detection in this sensitive field.

Random Forest and XGBoost had good accuracy since their ensemble mechanisms can represent constructive relationships between behavioral and psychological variables. Their high predictive performance was since they could reduce overfitting and learn nonlinear patterns, as well as appropriate preprocessing.

To make the findings stronger, it should be emphasized that the high functionality of the Random Forest and XGBoost shows that the ensemble models are capable of identifying the complex behavioral and psychological patterns underlying the risk of suicide. The fact that they have better recall and F1-scores implies that these models can better detect real high-risk patients, which is important in mental-health applications when the cost of failure to detect a vulnerable patient can be catastrophic. The findings not only indicate statistical superiority but also have a practical implication in the clinical settings in early screening and intervention. With the responsible use of such predictive tools and professional judgment, the latter can assist mental-health workers in the prioritization of at-risk individuals and the effectiveness of the suicide prevention process in general.

1.7 CONCLUSION

Suicidal acts are a significant public health issue. Every year, estimated one million people kill themselves worldwide. Globally, lowering suicidal actions (suicide attempts and deaths) is a significant public health problem. This paper presents a comprehensive framework for behavioral suicide risk prediction using four machine learning models. XGBoost was the best-performing model.

1.8 FUTURE WORKS

Subsequent research should be aimed to combine the proposed framework of suicide risk prediction with real-time monitoring solutions, extend it to the multi-modal data sources, including wearable data and conversation data, and facilitate personalisation of the developed models to the behavioural histories. Privacy-preserving methods like federated learning should also be put in the focus, as they allow sharing the data among the institutions safely, and strict verification must be conducted in distinct cultural and demographic settings. As another step, development of clinician-friendly decision support tools or clinician dashboards including human-in-the-loop feedback, longitudinal studies to quantify real-world effectiveness, and development of tools will aid translation to move closer to the target of matching the accuracy of the predictive tools with practical implementation. Lastly, we will be vital to set up sound ethical standards and governance to ascertain safe, fair and responsible application of the AI in the delicate mental health applications.

1.9 ETHICAL IMPLICATIONS

Suicide risk prediction through AI is a sensitive matter of mental-health data, thus solid ethical protection is needed. To maintain patient anonymity, all the records of this study were completely anonymized, and care was also taken to reduce the possibility of bias due to demographic imbalance. False negatives in suicide prediction may cause serious outcomes, and thus making use of accuracy alone is not safe, and, thus, recall and F1-score are of importance to be used in this paper in order to identify the true high-risk cases more effectively. Clinical judgment must not be replaced by AI predictions, and a human-in-the-loop solution is required to be able to make responsible decisions. In general, the ethical concerns of privacy protection, fairness, transparency, and safe deployment are essential in the application of AI in such risky areas.

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